

Central limit theorem

Central Limit Theorem: If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2 , then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

as $n \rightarrow \infty$, is the standard normal distribution $n(z; 0, 1)$.

Central Limit Theorem (Liapounoff's Form)

If $X_1, X_2, \dots, X_n, \dots$, be a sequence of independent RVs with $E(X_i) = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2, i = 1, 2, \dots$, and if $S_n = X_1 + X_2 + \dots + X_n$, then under certain general conditions,

S_n follows a normal distribution with mean $\mu = \sum_{i=1}^n \mu_i$ and variance $\sigma^2 = \sum_{i=1}^n \sigma_i^2$

as n tends to infinity.

Central Limit Theorem (Lindeberg-Levy's Form)

If $X_1, X_2, \dots, X_n, \dots$, be a sequence of independent identically distributed RVs with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2, i = 1, 2, \dots$, and if $S_n = X_1 + X_2 + \dots + X_n$, then under certain general conditions, S_n follows a normal distribution with mean $n\mu$ and variance $n\sigma^2$ as n tends to infinity.

Corollary

If $\bar{X} = \frac{1}{n} (X_1 + X_2 + \dots + X_n)$, then $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \frac{1}{n^2} (n \sigma^2) = \frac{\sigma^2}{n}$

$\therefore \bar{X}$ follows $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ as $n \rightarrow \infty$

Q. The lifetime of a certain brand of electric bulb may be considered a random variable with mean 1200 h and standard deviation 250 h. Find the probability using the Central Limit Theorem, that the average lifetime of 60 bulbs exceeds 1250 h. Given that $P(Z \geq 1.55) = 0.0606$.

Q2. If $X_1, X_2, \dots, X_n$ are independent Poisson variates with parameter 2, if $S_n = X_1 + X_2 + \dots + X_n$ and $n = 75$ then Use Central Limit Theorem to estimate $P(120 \leq S_n \leq 160)$.

Q3. The guaranteed average life of a certain type of electric light bulb is 1000 h with a standard deviation of 125 h. It is decided to sample the output so as to ensure that 90% of the bulbs do not fall short of guaranteed average by more than 2.5%. Use CLT to find the minimum sample size.

Practice problem

Q. If $X_1, X_2, X_3 \dots X_n$ are Poisson variates with parameter $\lambda = 4$, use the Central Limit Theorem to find $P(90 \leq S_n \leq 140)$ where $S_n = X_1 + X_2 + X_3 + \dots + X_n$ and 'n' = 50.

Q. The lifetime of a certain electrical commodity may be considered a random variable with mean 1100 h and standard deviation 225 h. Find the probability using the Central Limit Theorem, that the average lifetime of 64 commodities exceeds 1200 h. Given that $P(-\infty \leq Z \leq 3.56) = 0.9998$.